Jasmin Kareem | PhD Candidate in Explainable AI for News Recommender Systems

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| **Website** | **Google Scholar** | **Github** |

Summary

I am currently a joint PhD student working on **Explainability in News Recommender Systems** at both the University of Amsterdam in the Information Retrieval Lab (supervised by Maarten de Rijke) and at Eindhoven University of Technology within the JADS department (supervised by Martijn Willemsen). I also work with the Dutch-Belgian media conglomerate **DPG Media** as part of the Responsible Media Lab, which is an industry-academia collaboration with the goal of developing more responsible and transparent news systems. I work in the space **between human-computer interaction and computer science** and have a variety of **interests and expertise** including mechanistic interpretability, counterfactual explanations, news recommender systems, empirical user studies and qualitative research.

Relevant Projects and Publications

Counterfactual Explanations for News Recommender Systems

Currently in progress

I am working on a method for counterfactual explanations that can better explain news recommender system architectures. Currently most methods were designed for sequential or collaborative filtering architectures. Aiming for SIGIR 2027.

Is it Explanation or Control? An Experimental Study of Counterfactual Explanations and User Control in News Recommender Systems

Currently under submission

Investigated whether counterfactual explanations and/or reading-history control in personalized news recommenders were better for users through two iterations of a 2x2 study design, first with N=70 and then N=258 participants.

Do We Care About Personalization and Explainability? An Interview Study with News Recommendation Engineers at RecSys 2026 Main track

September 2026

Conducted qualitative interviews with news recommendation engineers to understand their perspectives on balancing personalization and explainability in practice, revealing gaps between academic priorities and industry constraints.

Education

Joint PhD Candidate at the Jheronimus Academy of Data Science (TU/e) and the University of Amsterdam

November 2023 - Present

My research is on interpretability in the context of news recommender systems. I am also affiliated with the University of Amsterdam as a member of the Information Retrieval Lab under the supervision of Maarten de Rijke and Martijn Willemsen. Expected graduation in November 2027.

MSc Data Science: Computer Science at Leiden University

September 2021 - November 2023

Thesis on using Algorithm Selection in MIP-based Neural Network Robustness Verification to improve efficiency. Supervised by Jan van Rijn and Holger Hoos. Average grade: 7.5

BSc Data Science at TU Eindhoven and Tilburg University

September 2017 - August 2021

Graduated with an average grade of 7.3. Thesis grade of 8 whilst interning at the Central Bureau of Statistics. Projects (Data Challenges) covered the topics in NLP and Time Series Forecasting.

Teaching Experience and Internships

Information Retrieval 2 at University of Amsterdam
November 2025 - Present
In this yearly course, I supervise groups of students on topics related to interpretability in information retrieval and recommender systems. The students reproduce recently published papers, and in some cases publish their findings in reproducibility tracks of Information Retrieval conferences.

Lecturer and TA for Master Course Explainable AI at Eindhoven University of Technology
February 2024 - Present
Created and taught 3 lectures + tutorials on feature attribution based explanations, counterfactual explanations, and on explanation visualizations.

Student Assistant for Master Course Social Network Analysis at Leiden University
September 2022 - January 2023
Attended tutorial sessions and answered questions from students regarding their assignments. I graded homework assignments and final project papers. Helped to give an online tutorials to PhD students at Kings College London (LISS238 Social Network Analysis)

Intern at Microsoft Netherlands, Customer Success Unit
April 2021 - April 2022
Between my Bachelor's and Master's studies, I completed an internship at Microsoft in the Netherlands, where I developed proof-of-concept applications for clients using Azure and Power Platform.

Volunteering work

Volunteer at European Summer School on Information Retrieval
July 2024
I was responsible for recording and collecting lectures by speakers and I organized one of the social events.

Organizer at the Dutch Belgian Workshop on Recommender Systems 2027 (DBWRS)
Present - April 2027
I am organizing the 2nd edition of DBWRS, together with others from the Netherlands. This event will take place in April 2027. Some of my current responsibilities include the website and developing the programme.

Mechanistic Interpretability Reading Group at the University of Amsterdam
2024 - Present
At the University of Amsterdam we have started and maintained a reading group aimed at keeping up with research on mechanistic interpretability (mostly interpreting and aligning LLMs). I regularly attend and have organized/chaired the reading group in 2025/2026. This reading group is attended by others working on the topic at the university and we sometimes invite guest speakers.

Skills

Programming languages: Python (primary), R, Julia, SQL, some C++

Specializations: Explainable AI (XAI), Recommender Systems, Information Retrieval, Human-Computer Interaction, Mechanistic Interpretability, Neural Network Verification, Automated Machine Learning (AutoML), Natural Language Processing (NLP)

Languages: English (fluent), German (fluent), Dutch (proficient)

Additional: Adaptable and quick learner; comfortable acquiring new languages and libraries as needed.